SIMATS ENGINEERING
Department of ComputerScience and Engineering ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	**Course Title:**	Computer Architecture for Multicore Systems
CO Assessed:	**CO1**	**Assessment:**	**ASSESSMENT TOOL 2-** CONCEPT MAPPING
Weightage:	**35%**	**Total Marks:**	**25**
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
Student Name:	**D.SRUJANA BAI**	**Reg.No.:**	**192571067**
Department:	**CS&BIOSCIENCE**	**Date:**	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

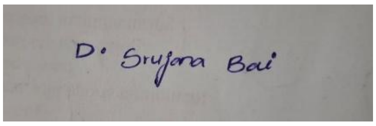
ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

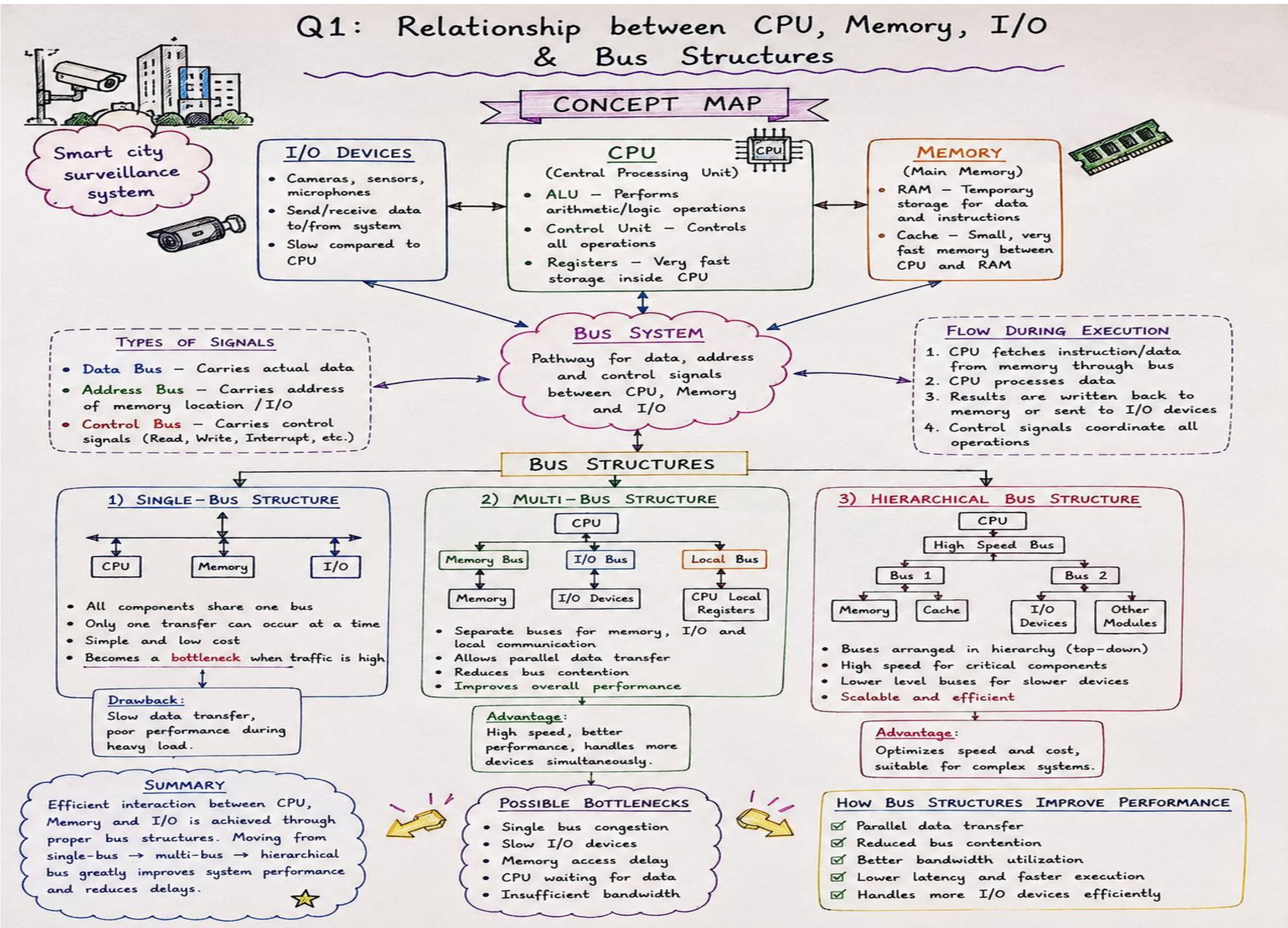
4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.

Code of Conduct:

I D.SRUJANA BAI(192571067)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

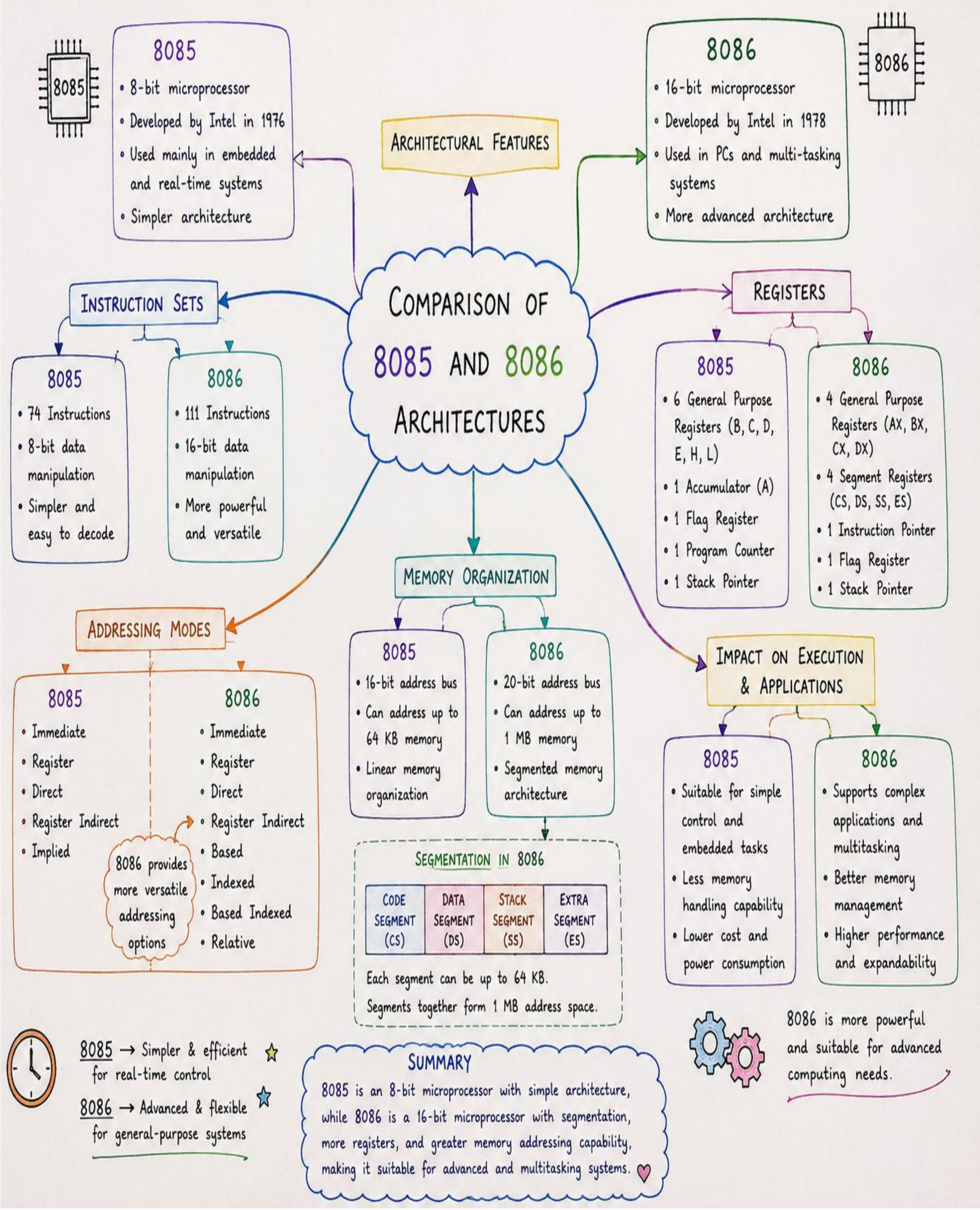


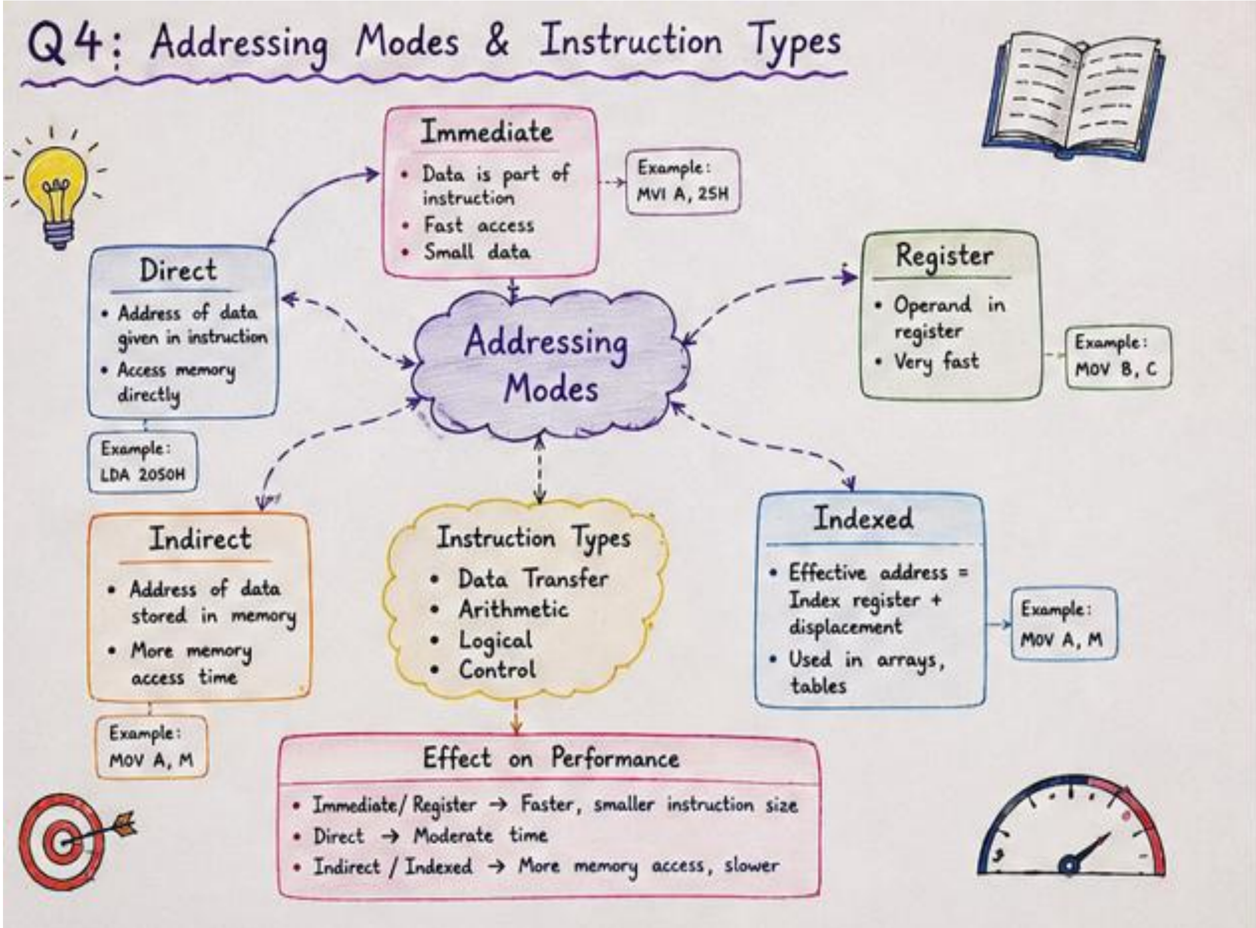
Signature of the student:

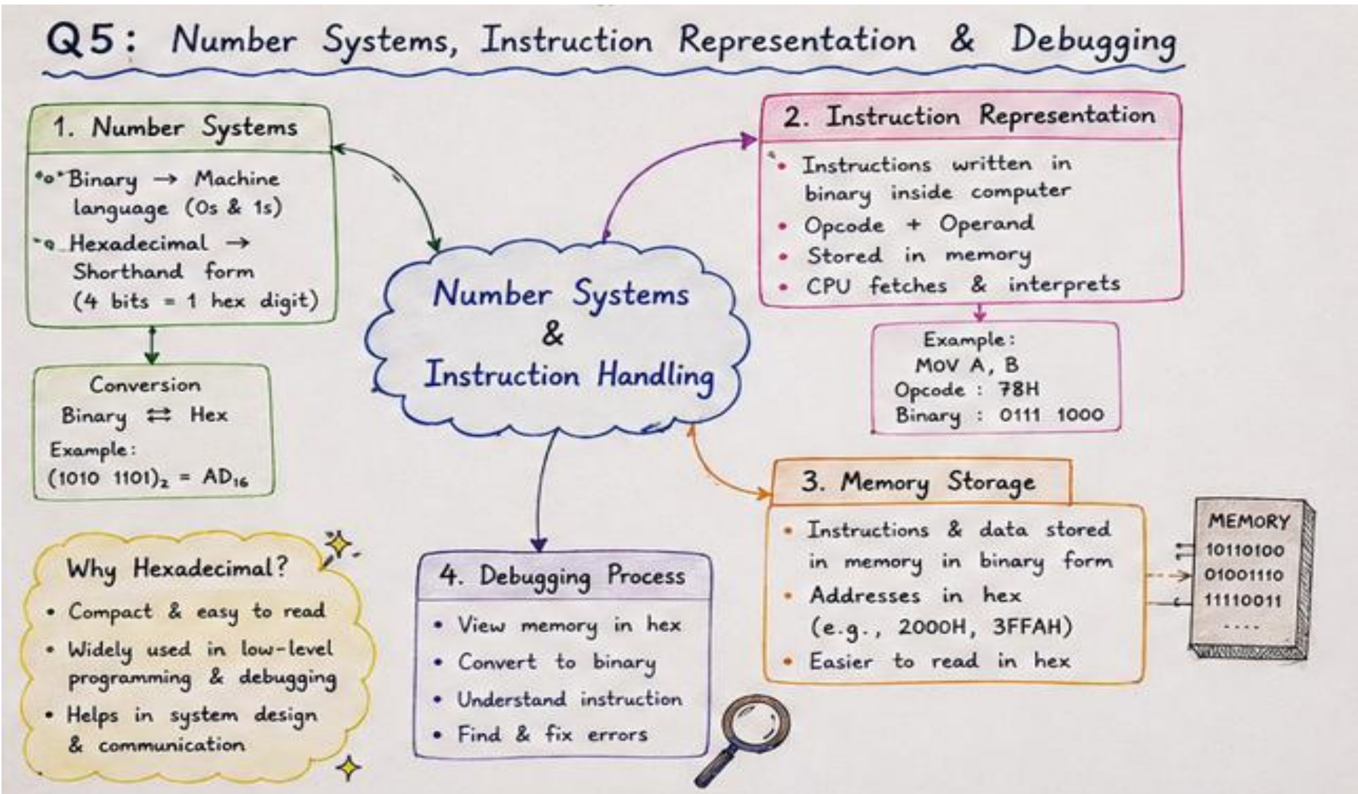


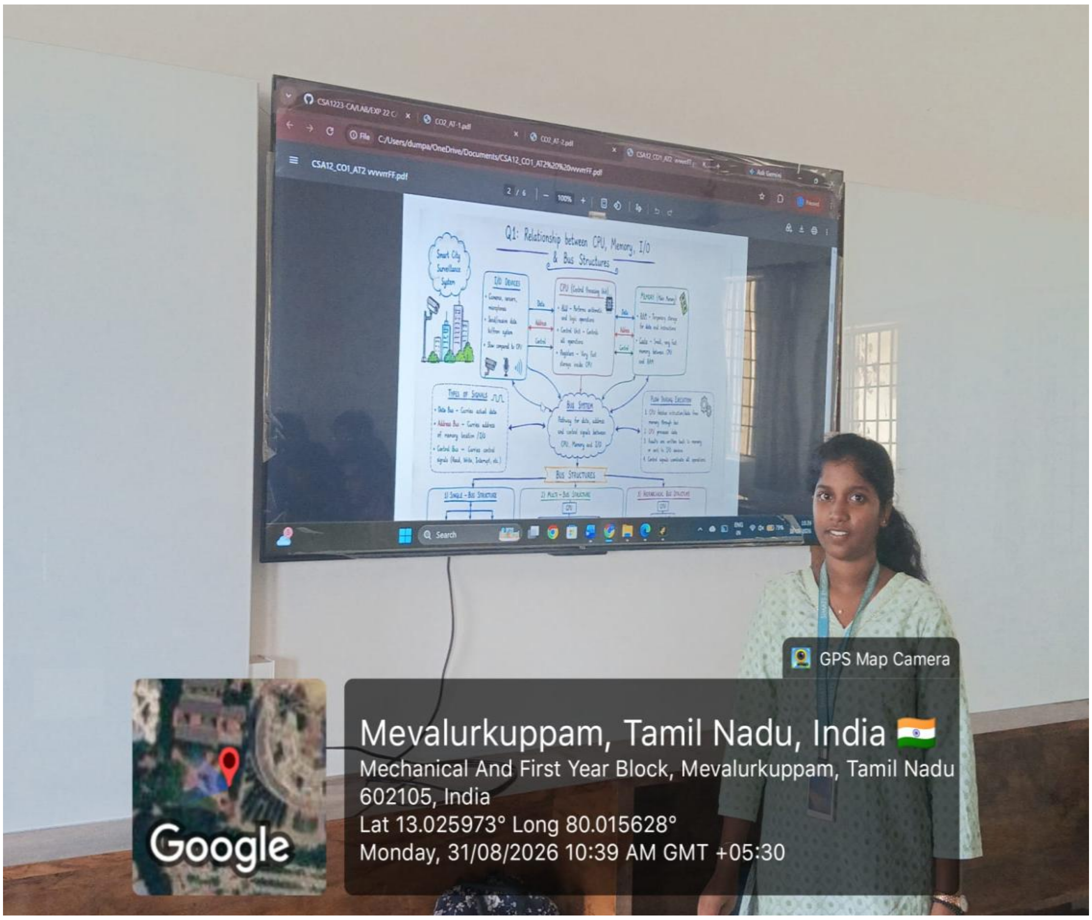
2Q.INSTRUCTION EXECUTION CYCLEAND CPU PERFORMANCE

3Q.8085&8086 ARCHITECTURES




Q4: Addressing Modes & Instruction Types
Addressing Modes
Immediate
• Data is part of instruction
• Fast access
• Small data
Example: MVI A, 25H
Direct
• Address of data given in instruction
• Access memory directly
Example: LDA 2050H
Register
• Operand in register
• Very fast
Example: MOV B, C
Indirect
• Address of data stored in memory
• More memory access time
Example: MOV A, M
Instruction Types
• Data Transfer
• Arithmetic
• Logical
• Control
Indexed
• Effective address = Index register + displacement
• Used in arrays, tables
Example: MOV A, M
Effect on Performance
• Immediate/ Register → Faster, smaller instruction size
• Direct → Moderate time
• Indirect / Indexed → More memory access, slower


Q5: Number Systems, Instruction Representation & Debugging
1. Number Systems
• Binary → Machine language (0s & 1s)
• Hexadecimal → Shorthand form (4 bits = 1 hex digit)
Conversion
Binary ⇄ Hex
Example:
(1010 1101)₂ = AD₁₆
Number Systems & Instruction Handling
2. Instruction Representation
• Instructions written in binary inside computer
• Opcode + Operand
• Stored in memory
• CPU fetches & interprets
Example:
MOV A, B
Opcode : 78H
Binary : 0111 1000
3. Memory Storage
• Instructions & data stored in memory in binary form
• Addresses in hex (e.g., 2000H, 3FFAH)
• Easier to read in hex
MEMORY
10110100
01001110
11110011
....
Why Hexadecimal?
• Compact & easy to read
• Widely used in low-level programming & debugging
• Helps in system design & communication
4. Debugging Process
• View memory in hex
• Convert to binary
• Understand instruction
• Find & fix errors



MARKS ALLOCATION:

Criteria	Q1			Q2			Q3			Q4			Q5		
	L3	L2	L1	L3	L2	L1	L3	L2	L1	L3	L2	L1	L3	L2	L1
Concept Identification (25%)															
Linkages (25%)															
Organization (20%)															
Integration of Literature (20%)															
Clarity (10%)															

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand